Multi-Agent Team Sentiment & Growth Feedback System

Strategy & Architecture Document

MVP Implementation

Owner: Engineering Manager — AI Platform

Version 1.0 | March 2025 | Status: MVP Prototype Implemented

1. Executive Strategy

Vision

Transform engineering team feedback from static annual surveys into a continuous, AI-assisted feedback intelligence system that converts team sentiment into measurable, actionable improvements for engineering managers.

Mission

Build an agentic AI platform that automates the feedback-to-action loop — enabling organizations to continuously measure and improve team health, reduce burnout risk, and accelerate career growth.

Problem Statement

Engineering Managers lack structured, recurring feedback loops to identify morale shifts, burnout signals, and career growth concerns. Annual surveys are static, delayed, and disconnected from real action. This results in:

- Reduced employee engagement and rising burnout
- Higher voluntary attrition with significant replacement cost
- Missed opportunities for proactive career development
- EM bandwidth wasted on reactive, not proactive, management

Why Now?

- Hybrid work has eroded informal feedback channels
- Burnout rates in tech are at historic highs
- LLM-based agentic systems are now mature enough for explainable automation
- Engineering Manager bandwidth is increasingly constrained

2. Strategic Pillars

Pillar	Description
Feedback as a Service	Automated, periodic pulse surveys covering learning, career growth, wellbeing, collaboration, and leadership — replacing infrequent, static annual surveys.
Agentic Intelligence	Specialized AI agents handle distinct stages of the feedback lifecycle with explainable outputs at each step.
Insights to Action	Raw survey responses are automatically converted into sentiment scores, theme extraction, risk identification, and prioritized manager action plans.
Privacy & Trust First	Aggregated analysis with anonymization thresholds and no individual response attribution protects employee trust.
Dynamic Adaptation	The system learns from prior survey cycles to refine future questions, suppress redundant prompts, and surface emerging themes.

3. Target Users & Value Proposition

Primary Users

- Engineering Managers (EMs): Weekly summaries, early-warning alerts, AI coaching suggestions, 1:1 conversation prompts
- Directors of Engineering: Team health trends, retention risk heatmaps, capability gap visibility

Secondary Users

- HR Business Partners: Organizational sentiment trends, attrition risk reports
- People Operations: Aggregate engagement data, benchmarking
- VP Engineering / CTO: Executive dashboards, strategic KPIs

Value Proposition

Stakeholder	Value Delivered
Engineering Managers	Data-informed decisions weekly, not annually. AI-generated 1:1 prompts and coaching suggestions save time and improve impact.
Team Members	Feel heard continuously. Feedback actually drives visible change rather than disappearing into an annual report.
People Ops / HR	Proactive attrition risk signals. Structured data to justify headcount and wellbeing investments.
Executive Leadership	Measurable culture KPIs. Reduced voluntary attrition and faster team capability growth.

4. MVP Product Scope

Implemented Capabilities

Capability	Description	Status
Pulse Survey Collection	Google Forms integration	☑ Implemented
AI Sentiment Analysis	LLM-based scoring per response set	☑ Implemented
Theme Detection	Automated key theme extraction	☑ Implemented
Risk Identification	Burnout and disengagement signal detection	☑ Implemented
AI Action Plan Generation	Manager-ready recommendations per cycle	☑ Implemented
Historical Insights Storage	Google Sheets-based trend data	☑ Implemented
Manager Dashboard	Google Sheets output view	🔧 Phase 2
Slack Alerts	Early-warning push notifications to EM	🔧 Phase 2
Predictive Modeling	Burnout forecasting & attrition risk scoring	🔧 Phase 3
HRIS Integration	Workday, BambooHR connectors	🔧 Phase 4

Out of Scope for MVP

- Deep root-cause analysis of burnout (planned Phase 2)
- Sprint-cycle survey alignment (planned Phase 2)
- Stakeholder expectation document ingestion (planned Phase 2)
- Voice-enabled feedback collection

5. System Trigger, Data Points & Autonomy Model

How the System is Triggered

Trigger Type	Description
Scheduled (MVP)	n8n cron trigger fires on a weekly or monthly cadence. Sends survey link to team via email or Slack.
Event-Driven (Phase 2)	Triggered after sprint completion, project milestones, or significant org events (reorg, layoffs).
On-Demand (Phase-3)	EM manually initiates a pulse check outside the regular cycle via dashboard.

Threshold-Based (Phase 3)	Predictive model detects early risk signals and auto-triggers a targeted check-in survey.

Key Data Points Collected

- Learning & skill development satisfaction (1–5 scale + open text)
- Career progression clarity (1–5 scale + open text)
- Work-life balance and wellbeing (1–5 scale + burnout risk indicators)
- Team collaboration & culture health
- Leadership support quality
- Open-ended: What is one thing that would most improve your team experience?

Level of Autonomy

The MVP operates at Level 2 (Supervised Automation) on the autonomy scale:

- Agents act autonomously to analyze, score, and generate action plans
- Managers review and approve before acting on recommendations – Deferred for future release
- High-risk alerts (burnout, attrition) trigger human-in-the-loop review – Deferred for future release
- Phase 3 will explore Level 3: proactive recommendations pushed without explicit EM trigger

6. System Architecture

Architecture Overview

The MVP is built on n8n for rapid iteration with a modular, agent-based design. The architecture separates concerns across four layers:

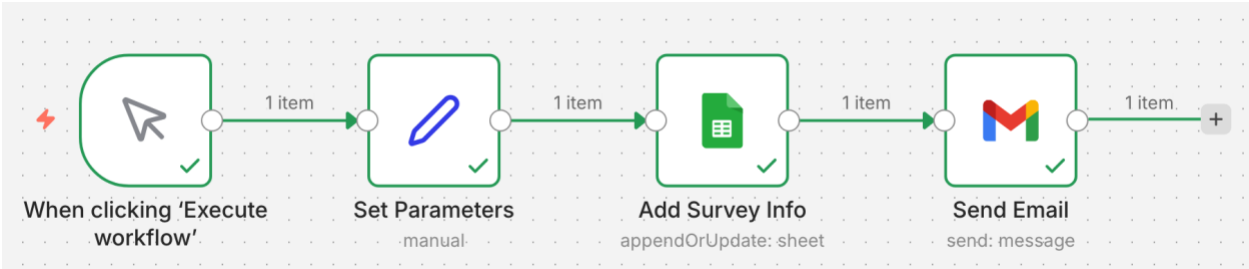
LAYER	COMPONENTS
Input Layer	Google Forms (MVP) → Phase 2: Slack, Email, Mobile
Data Store Layer	Google Sheets: Response Store, Insights Store, Action Plan Store → Phase 2+: PostgreSQL / Pinecone
Orchestration Layer	n8n Workflow Engine → Cron trigger → Agent pipeline coordination
AI Agent Layer	MVP: Linear, Phase-2: Survey Agent | Insight Agent | Action Agent | Orchestration Agent
Delivery Layer	Google Sheets Dashboard (MVP) → Phase 2: Slack alerts, email digests, web dashboard

Architecture Flow Diagram (Text Representation)

```
[ Google Forms (Survey Input) ]
          ↓  (response storage)
[ Google Sheets — Response Store ]
          ↓  (n8n scheduled trigger)
[ n8n Orchestration Agent ]
```

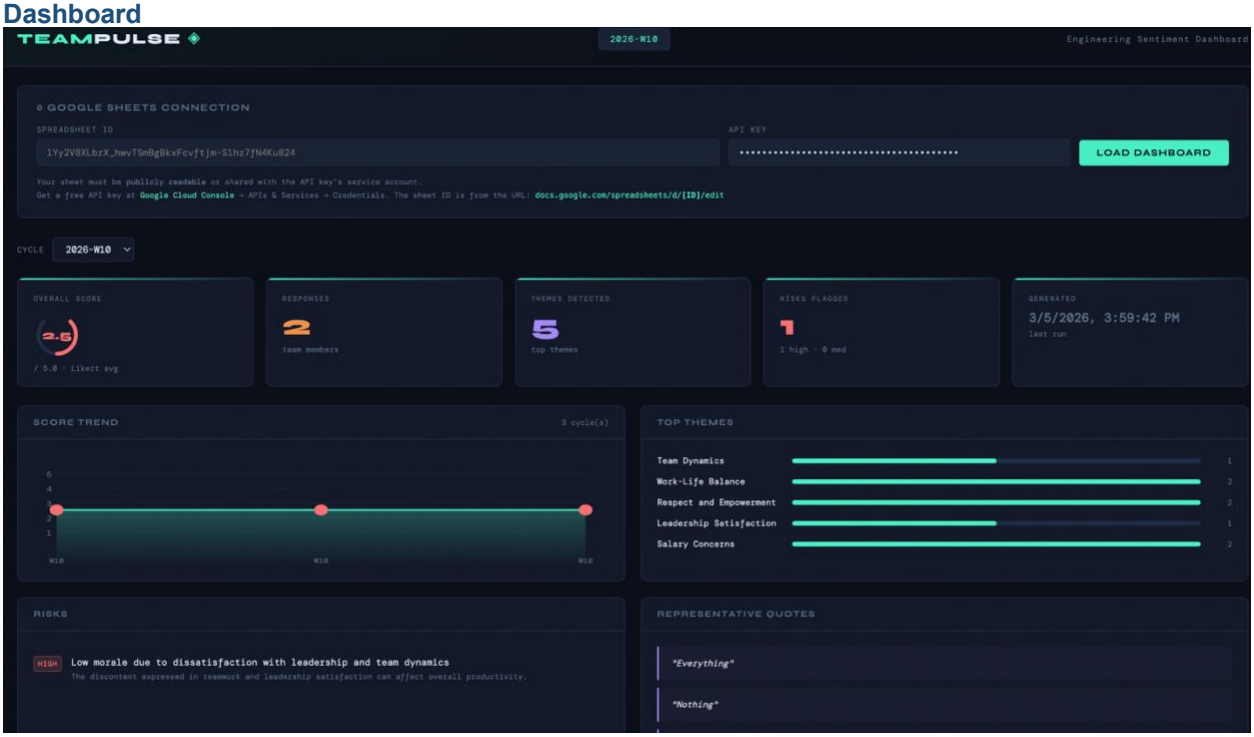
```
      ├──→ [ Survey Agent ]  →  Formats & validates responses
      ├──→ [ Insight Agent ]  →  Sentiment + Themes + Risk
      └──→ [ Action Agent ]  →  Prioritized recommendations
           ↓  (write outputs)
[ Google Sheets — Insights Store ]  +  [ Action Plan Store ]
           ↓  (manager review)
[ EM Dashboard (Sheets) ]
           ↓  (Phase 2)
[ Slack Alerts ]  [ Email Digest ]  [ Web Dashboard ]
```

Survey Sender Flow – Cron



Team Sentiment Analysis and Action Planner - Cron

Dashboard



7. Agentic Workflow Design

Agent Breakdown

Agent	Technology	Responsibility
Orchestration Agent	n8n cron + coordinator	Triggers pipeline, sequences agents, handles retries and error routing
Survey Agent	n8n + Google Forms API	Validates and retrieves new responses, aggregates into structured dataset
Insight Agent	OpenAI GPT + structured prompts	Produces sentiment score (1–10), extracts top 3–5 themes, flags risk signals
Action Agent	OpenAI GPT + structured prompts	Generates 3–5 prioritized manager actions with associated KPIs
Tracking Agent (Phase 2)	n8n + Sheets/HRIS	Monitors action plan completion, closes the feedback loop
Wellbeing Risk Agent (Phase 3)	ML model + LLM	Detects burnout patterns across multiple survey cycles
Growth Gap Agent (Phase 3)	LLM + skills taxonomy	Identifies skill gaps and maps to learning recommendations

Agent Interaction & Guardrails

- All agent outputs are grounded in retrieved response data — no hallucination of survey answers
- Insight Agent confidence score < 0.7 falls back to rule-based theme extraction
- Action Agent outputs are labeled as AI-suggested and require EM review before action
- Anonymization threshold: analysis suppressed if fewer than 5 responses in a cycle
- Bias detection pass runs on sentiment outputs before storage

8. End-to-End Workflow

Step	Stage	Description
Step 1	Survey Trigger	n8n cron fires. Survey Agent sends Google Forms link to team members via email/Slack.
Step 2	Response Collection	Team submits responses. Google Sheets auto-collects all submissions.
Step 3	Aggregation	n8n retrieves new responses. Survey Agent validates and structures the dataset.
Step 4	AI Analysis	Insight Agent analyzes aggregate data: sentiment score, top themes, risk flags.

Step 5	Insights Storage	Structured insights written to Insights Store in Google Sheets with cycle metadata.
Step 6	Action Generation	Action Agent generates 3–5 prioritized manager recommendations with KPIs.
Step 7	Action Storage	Action plans written to Action Plan Store with target dates and owners.
Step 8	EM Review	Manager accesses Sheets dashboard to review insights and action plans.
Step 9 (Phase 2)	Delivery & Alerts	Slack message delivers weekly summary. High-risk alerts sent immediately.
Step 10 (Phase 2)	Tracking	Tracking Agent monitors completion of prior action plans each new cycle.

9. Infrastructure & AI Stack

MVP Stack

Component	Technology / Notes
Survey Layer	Google Forms — zero-infrastructure survey collection
Data Store	Google Sheets — response store, insights store, action plan store
Workflow Orchestration	n8n Cloud — cron scheduling, agent coordination, API integrations
AI Model	OpenAI GPT-4o mini — sentiment analysis, theme extraction, action generation
Processing Logic	JavaScript nodes inside n8n for data transformation
Dashboard	HTML dashboard for EM consumption

Future Production Stack (Phase 3+)

- Deployment: AWS Bedrock for scalability and enterprise compliance
- Orchestration: Migration from n8n to Kubernetes-based microservices for scale
- Data Store: PostgreSQL for structured data; Pinecone for vector embeddings (user trend memory)
- Messaging: Slack API, MS Teams webhooks for real-time EM notifications
- HRIS Integration: Workday, BambooHR connectors for org-level enrichment
- Monitoring: Datadog or AWS CloudWatch for pipeline health; custom LLM quality dashboards

10. Privacy & Responsible AI

Control	Description
Anonymization	No individual response attribution is surfaced to managers. Analysis is always aggregate.
Threshold Guard	Analysis is suppressed if fewer than 5 responses are received in a cycle to prevent de-anonymization.
PII Redaction	Survey Agent strips any accidentally submitted PII before passing data to Insight Agent.
Human-in-the-Loop	High-risk alerts (burnout, attrition) require EM review before any action is taken.
AI Advisory Only	All outputs are clearly labeled as AI-suggested. Managers retain full decision authority.
Bias Detection	Sentiment scoring outputs are reviewed for demographic or role-based bias patterns.
Data Retention	Response data retained for 12 months by default; configurable per org policy.

11. Risk Management

Risk	Likelihood	Mitigation
Survey Fatigue	High	Adaptive question reduction; limit to 5–8 questions per cycle. Vary question sets.
Manager Inaction	High	Automated reminders; KPI tracking on action plan completion rates.
AI Hallucination	Medium	Retrieval-grounded prompts; confidence scoring; rule-based fallback.
Privacy Breach	Medium	Anonymization thresholds; PII redaction; enterprise-grade encryption.
Low Response Rate	Medium	Slack/email nudges; EM visibility into participation (not individual answers).
Over-reliance on Score	Low	Contextual narrative always accompanies scores; trend comparison required.
Vendor Lock-in (n8n)	Low	Architecture designed for portability; core logic in JS nodes, not n8n-specific DSL.

12. Success Metrics & KPIs

MVP Evaluation Criteria

Metric	Target	Measurement Method
Survey Response Rate	> 60% per cycle	Google Sheets dashboard
Action Plan Quality	Manager usefulness rating > 4/5	Post-cycle EM survey
Insight Accuracy	Human review validation > 80%	Manual spot-check per cycle
EM Adoption	> 3 EMs completing full pilot	Pilot participation tracking
Cycle Completion Time	< 5 min from trigger to dashboard	n8n execution log

Phase 2+ Metrics

- Engagement uplift vs. baseline: target +10% in 6 months
- Voluntary attrition reduction: target -15% in 12 months
- Burnout risk early detection rate: flag > 80% of high-risk cases before they escalate
- Internal mobility increase: +1 internal transfer per 10-person team per year
- Action plan completion rate: > 70% of generated plans completed within 60 days

13. Product Roadmap

Phase	Status	Key Deliverables
MVP (Now)	Completed	Google Forms surveys, LLM sentiment analysis, theme detection, risk ID, action plans, EM dashboard UI
Phase 2 (Q3–Q4 2025)	In Planning	Slack delivery, trend tracking, sprint-cycle alignment, deep-dive burnout root cause, stakeholder doc ingestion
Phase 3 (Q1–Q2 2026)	Future Vision	Predictive attrition model, promotion readiness signals, personalized career growth engine, voice feedback
Phase 4 (2026+)	Future Vision	HRIS integration (Workday, BambooHR), LMS integration, enterprise compliance, data warehouse analytics, white-label SaaS

14. Resource Asks & Budget Justification

Headcount

Stage	Role	Justification
Now — Prototype	EM (Internal)	EM self-building prototype using n8n and OpenAI APIs. No additional headcount.
MVP → Production	0.5 FTE Engineer	EM reprioritizes an existing engineer part-time to harden the MVP, add Slack integration, and begin Phase 2 features.
Phase 2–3 Scale	ML/AI Engineer	Validates LLM quality, builds feedback loops, implements confidence scoring and bias detection.
Phase 2–3 Scale	Full-Stack Engineer	Builds EM-facing dashboard UI, Slack bot, and notification pipelines.
Phase 3+	Data/Infra Engineer	Deploys production infrastructure, HRIS integrations, monitoring, and scalability.

Budget Request: MVP to Production

Item	Estimated Cost	Notes
AI Model APIs (OpenAI)	$1,000–$2,000/mo	Prototype usage ~1M tokens/month. Scale to GPT-4o mini in production to reduce cost.
n8n Cloud	$200–$500/mo	Workflow orchestration. Self-hosting on AWS reduces cost at scale.
Cloud Infrastructure	$500–$1,000/mo	App hosting, DB, storage, API gateways (AWS or GCP).
Monitoring & CI/CD	$300–$500/mo	Datadog or CloudWatch; GitHub Actions for deployment pipelines.
Total (Monthly)	~$2,000–$4,000/mo	Total monthly operating cost at MVP-to-Production scale.
Total (Annual)	~$24K–$48K/year	Annual run rate. Additional one-time setup costs ~$5K–$10K.

NOTE	The n8n prototype can be demonstrated at near-zero cost using the free tier. Budget above is for hardened, production-grade deployment serving 10–50 engineering teams.

15. Appendix: Design Patterns & Production Decisions

Design Patterns Used/Imagined

- Sequential: For simplicity and MVP, current design is based on sequential approach where each component in the flow is tightly coupled with other component
- Orchestrator-Worker: Orchestration Agent decomposes the workflow and delegates to specialist agents

- RAG (Retrieval-Augmented Generation) — Phase 2: Historical survey data used to ground LLM analysis and prevent hallucination
- Event-Driven Architecture: Cron + webhook triggers initiate pipeline; future sprint-cycle events will drive dynamic triggers
- Human-in-the-Loop: High-risk recommendations require EM review before actioning
- Separation of Concerns: Each agent has a single, well-defined responsibility

Productionization Plan

- Phase 1: n8n Cloud (current) — zero-ops, fast iteration
- Phase 2: Rearchitect to implement different agents for different tasks, make it modular/configuration based so tools and LLM model can be replaced without significant cost and risk. Deploy to Amazon Bedrock
- Phase 3: Make the Survey adaptive, predictive based on previous surveys and market trends
- Phase 4: Evaluate the performance, compliance and future scalability needs, and change the deployment model or migrate to custom microservices.
- Phase 5: Integrate with HRIS
- Phase 6: Cost Optimization: Evaluate GPT-4o mini and replace with alternative cost effective LLM models

Monitoring & Observability

- Pipeline health: n8n execution logs + CloudWatch/Datadog alerts for failures
- LLM quality: Spot-check output quality every cycle; confidence score dashboards
- Survey health: Response rate trend monitoring; automated EM nudge if rate drops below 50%
- Data quality: Schema validation on Google Sheets ingestion; alert on malformed responses